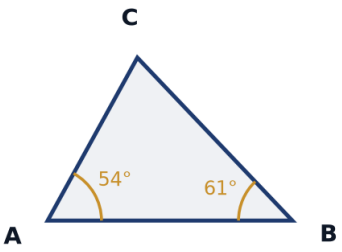
Solving for sides in right triangles

Core Concepts: Solving for sides in right triangles

1 In $\triangle ABC$ studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 85°

b. 95°

c. 77°

d. 130°

2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 1$?

a. 6

b. 11

c. 9

d. 12

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 In $\triangle ABC$ studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 78°

b. 88°

c. 70°

d. 130°

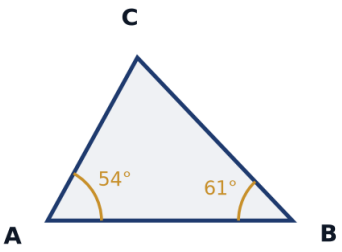
5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm

c. 16 cm

d. 36 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 2$?

a. 9

b. 14

c. 12

d. 15

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 71°

b. 81°

c. 63°

d. 130°

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

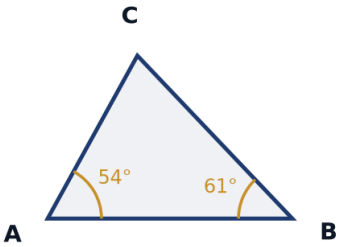
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?

a. 12

b. 17

c. 15

d. 18



Exam-Style Problem Solving & Modeling

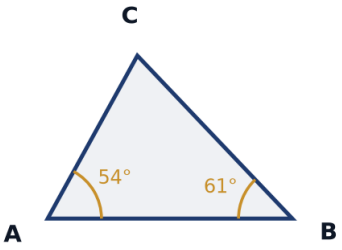
10 In $\triangle ABC$ studying solving for sides in right triangles, $m\angle A = 50^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 64°

b. 74°

c. 56°

d. 130°



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 15

b. 20

c. 18

d. 21

© ClipSAT: By Mr. Mohamed Abdallah +966597688647